GHCAA Requirements Catalogue

57 functional requirements, 37 non-functional requirements and 11 quality-attribute scenarios. Printed 2026-10-31.

Summary

Functional area	Requirements	Must	Should	Could
Membership, Registration and Profile	8	6	2	0
Authentication, OTP and Session Management	5	4	1	0
Events, Registration, Attendance and Gallery	7	2	3	2
Payments, Dues and Financial Records	7	7	0	0
News, Notices and Communication	8	3	4	1
Governance	9	8	0	1
Administration, Configuration and Site Content	7	5	1	1
Mobile Application Requirements	5	3	1	1
Job Board	1	0	0	1
Total	**57**	**38**	**12**	**7**

Quality characteristic (ISO/IEC 25010:2011)	Requirements
Functional suitability	2
Performance efficiency	5
Compatibility	3
Usability	5
Reliability	6
Security	9
Maintainability	4
Portability	3
Total	**37**

Priority: M must, S should, C could. Source: D document analysis, with the constitution article cited where it applies; I interview; O observation; C competitor analysis; R dated stakeholder request.

Functional requirements

Membership, Registration and Profile

ID	Requirement	Source	Pri
FR-01	The system shall provide a public multi-step registration wizard that captures personal, academic and professional particulars, permits an applicant to save progress and resume, and validates each step before advancing.	I, C	M

ID	Requirement	Source	Pri
FR-02	The system shall create an application in status *Applied* on submission, record the submission timestamp, and expose the application state to the applicant through a status view. Statuses shall be limited to Applied, Active, Inactive by non-payment, Inactive by resignation, Terminated and Rejected.	D (Art. III §E), I	M
FR-03	The system shall allow a member to maintain his own profile and to control, per field, whether that field is disclosed in the directory, with non-disclosure as the default for contact and identity fields.	I, D	M
FR-04	The system shall require at least one academic record at Govt. Haraganga College before an application may be submitted, and shall permit any number of additional academic and professional records.	D (Art. III), O	M
FR-05	The system shall provide a directory searchable by name, batch, academic session, subject, profession and location, returning only fields the owning member has disclosed, except to administrative users.	I, C	M
FR-06	The system shall issue each active member a digital identity card bearing the membership number, photograph, tier and a machine-readable code that resolves to a verification endpoint.	I, C	S
FR-07	The system shall present each member his own standing, comprising membership number, tier, status, dues raised, dues settled and outstanding balance.	I, O	M
FR-55	The system shall identify the institution-defining academic record through a flag evaluated against the active organisation profile, rather than a name specific to one institution, preserving the existing true/false meaning and the first-record rule of FR-04, and shall accept the field's legacy name from a client not yet updated, mapping it to the same stored value.	R	S

Authentication, OTP and Session Management

ID	Requirement	Source	Pri
FR-08	The system shall authenticate members by membership number or registered email address together with a password, storing passwords only as an adaptive salted hash.	I	M
FR-09	The system shall verify a registered email address and mobile number by a single-use, time-limited code, and shall reset a forgotten password by a single-use, time-limited link.	I	M
FR-10	The system shall permit sign-in by a federated identity provider, associating the federated identity with exactly one member record and requiring completion of the member profile before portal access is granted.	I, C	S
FR-11	The system shall issue a short-lived access token and a rotating refresh token, shall support explicit sign-out, and shall invalidate all outstanding sessions for a member when his credentials, roles or status change.	D, I	M
FR-12	The system shall prevent a member whose mandatory profile particulars are incomplete from using member functions other than profile completion, and shall indicate what remains outstanding.	O	M

Events, Registration, Attendance and Gallery

ID	Requirement	Source	Pri
FR-13	The system shall allow an administrator to publish an event with title, description, venue, start and end, capacity, fee and registration deadline, and to amend or cancel it with notification to registrants.	I	M
FR-14	The system shall allow a member to register for a published event, subject to capacity, deadline and any fee condition, and to cancel his registration.	I	M

ID	Requirement	Source	Pri
FR-15	The system shall place registrations beyond capacity on an ordered waiting list and shall promote from the head of that list, in order, when a place is released.	C	S
FR-16	The system shall record attendance by scanning the member's identity code at the venue, and shall reject a code that does not correspond to a valid registration for that event.	I, C	S
FR-17	The system shall maintain a per-event photograph gallery, uploaded by administrators, visible according to the event's publication setting.	I	C
FR-18	The system shall report per event the registered count, the waitlisted count, the attended count and the fee collected.	I	S
FR-53	The system shall permit a member to submit a photograph or album to an event's gallery, and shall hold the submission unpublished until an administrator approves or rejects it, with a recorded reason on rejection.	R	C

Payments, Dues and Financial Records

ID	Requirement	Source	Pri
FR-19	The system shall raise the admission fee and the first annual subscription as ledger obligations at the moment a membership is approved, at the rates then configured.	D (Art. VI), O	M
FR-20	The system shall raise the annual subscription for each active member on the anniversary configured for the purpose, and shall not raise it for members whose tier is exempt.	D (Art. III, Art. VI)	M
FR-21	The system shall record, for every payment, the amount, the date, the channel, the member concerned, the transaction reference supplied by the member, the uploaded evidence, the verifying officer and the verification timestamp.	O	M
FR-22	The system shall permit a member to declare a payment made through an external channel by supplying a transaction reference and uploading evidence, and shall hold that declaration unverified until an officer acts on it.	I, D	M
FR-23	The system shall present administrators a queue of unverified payment declarations and of pending applications, ordered by age, and shall indicate any item older than thirty days.	D (Art. III §E), O	M
FR-24	The system shall compute each member's standing from raised and settled obligations, shall flag arrears, and shall require an administrative act, with a recorded reason, for any change of membership status arising from arrears.	D (Art. X), I	M
FR-25	The system shall issue a downloadable receipt bearing the membership number, the amount, the channel, the reference and the verifying officer, for every verified payment, and shall not permit a verified financial record to be amended or deleted; corrections shall be made by a compensating entry.	D (Art. VI), I	M

News, Notices and Communication

ID	Requirement	Source	Pri
FR-26	The system shall maintain posts of two kinds, news and notice, with a common structure and differing authorship rules.	D (Art. IV), I	M
FR-27	The system shall permit members to submit news items for editorial approval, and shall refuse any member submission of kind notice; notices shall be creatable only by an authorised officer.	D (Art. IV), I	M
FR-28	The system shall permit an officer to publish, unpublish, pin, edit and archive posts, retaining the archived record rather than deleting it.	I	M

ID	Requirement	Source	Pri
FR-29	The system shall permit an officer to compose and dispatch an email or SMS communication to a selected segment of the membership, where a segment may be defined by tier, status, batch or event participation, and shall record the channel, recipient member, delivery status, scope and time.	C, I	S
FR-29a	The system shall let a member view only the member's own paginated outbound communication history, including channel, status, scope, message detail and delivery time.	I	S
FR-29b	The system shall record unavailable and failed provider outcomes without exposing failure detail on successful history rows.	I	S
FR-30	The system shall deliver in-application notifications for events material to a member, including application decision, payment verification, event promotion from the waiting list, and notices.	I	S
FR-31	The system shall provide member-to-member and member-to-officer messaging in real time, subject to the privacy settings of FR-03.	I, C	C

Governance

ID	Requirement	Source	Pri
FR-32	The system shall publish the constitution to members and to the public, presenting always the version in force by effective date, and shall not require any page or link to name a version.	D (Art. VIII), I	M
FR-33	The system shall retain every ratified version of the constitution with its version identifier, effective date and summary of change, shall present that history, and shall supersede rather than remove a version when a later one takes effect.	D (Art. VIII)	M
FR-34	The system shall maintain the Executive Committee record, comprising the fifteen positions, the holder of each, the term start and end, and whether the position is elected or ex officio, and shall retain past committees.	D (Art. IV)	M
FR-35	The system shall permit an authorised officer to open an amendment proposal for voting, recording the proposing route, the circulation date and the closing date, and shall enforce a minimum circulation period of fourteen days before the vote opens.	D (Art. VIII)	M
FR-36	The system shall admit a vote on an amendment proposal only from a member who is a voting member in good standing at the time of voting, shall admit exactly one vote per eligible member, and shall record the eligibility determination together with the vote.	D (Art. III §B, Art. V)	M
FR-37	The system shall report an amendment vote as the count of votes cast for and against, the number of eligible voters, and whether the two-thirds threshold was met, and shall not itself declare the amendment adopted.	D (Art. VIII)	M
FR-38	The system shall maintain the roll of members eligible to vote at a forthcoming election, derived from tier and standing, and shall make that roll available to the Election Commission in a form that can be published and challenged.	D (Art. V), election regulations	M
FR-39	The system shall support a persisted election workflow for authorized administration, including a frozen voter roll, nominations, secret ballots, counting and declaration, while detailed legal form wording and institutional adoption remain governed by the Election Commission's election documents.	D (Art. V), election regulations, I	M
FR-40	The system shall support non-binding member polls, open to a configurable eligibility set, reporting counts only.	I	C

Administration, Configuration and Site Content

ID	Requirement	Source	Pri
FR-41	The system shall provide an administrative console covering membership, payments, events, posts, communication, governance and configuration, with functions available according to the acting user's role.	I	M
FR-42	The system shall hold operational parameters as configuration rather than in code, including fee amounts, payment channel particulars and instructions, the subscription anniversary, rate limits and contact details, and shall permit an authorised administrator to change them without redeployment.	I, O	M
FR-43	The system shall record every administrative act that changes a member's status, verifies a payment, or amends configuration, together with the acting user and the time.	O, D (Art. VI)	M
FR-44	The system shall implement archival deletion, such that a record marked archived is excluded from ordinary retrieval while remaining recoverable, and shall not permit destructive deletion of membership or financial records through the interface.	D (Art. VI), I	M
FR-45	The system shall permit an administrator to edit the text and images of public informational pages without redeployment.	I	S
FR-46	The system shall present a public site giving the Association's purpose, its office holders, published news and notices, the constitution, election information, a gallery, and a route to apply for membership.	I, C	M
FR-47	The system shall permit an administrator to select the presentation theme from a configured set, applying to the public site and to the member portal.	I	C

Mobile Application Requirements

ID	Requirement	Source	Pri
FR-48	The mobile application shall provide sign-in, profile maintenance, the directory, events, the identity card, payment declaration and notices, from a single codebase for both major platforms.	I, C	M
FR-49	The mobile application shall render the identity card and its machine-readable code without a network connection once the member has signed in at least once.	I	S
FR-50	The mobile application shall capture and compress a photograph or evidence image on the device before upload, to the size limit stated in NFR-P4.	O	M
FR-51	The mobile application shall discourage capture of the identity card and profile screens by screen recording or screenshot where the platform provides the means.	I	C
FR-52	The mobile application shall report an unavailable or failed network operation to the member in terms of what to do next, and shall not lose entered data on failure.	O	M

Job Board

ID	Requirement	Source	Pri
FR-54	The system shall permit a member to post a job or career opportunity, and shall hold the posting unpublished until an administrator approves or rejects it, with a recorded reason on rejection.	R	C

Non-functional requirements

Each requirement has a measurable acceptance criterion. A requirement that could not be given one was rejected rather than weakened.

Functional suitability

ID	Requirement and acceptance criterion
NFR-F1	Every functional requirement of priority M shall have at least one automated test asserting its principal behaviour. Criterion: the traceability matrix leaves no M-priority requirement unmapped.
NFR-F2	No enforced constitutional rule shall exist in the implementation without a recorded domain constraint, and no must-have domain constraint shall be unimplemented. Criterion: traceability from constraint to code is complete in both directions.

Performance efficiency

ID	Requirement and acceptance criterion
NFR-P1	Read endpoints shall respond within 500 ms at the 95th percentile under a load of 50 concurrent users on the production tier.
NFR-P2	Write endpoints shall respond within 1,000 ms at the 95th percentile under the same load.
NFR-P3	The web client's initial route shall become interactive within 3 s on a mid-range Android handset over a 3G-class connection, with an initial transfer budget of 500 KB compressed.
NFR-P4	Uploaded images shall be reduced to at most 350 KB before storage, without visible loss at the sizes at which they are displayed.
NFR-P5	Directory search over the full membership shall return the first page within 800 ms at the 95th percentile.

Compatibility

ID	Requirement and acceptance criterion
NFR-C1	The web client shall function on the current and preceding major versions of Chrome, Firefox, Safari and Edge.
NFR-C2	The mobile application shall support Android 8.0 and above and iOS 13 and above.
NFR-C3	The persistence layer shall be provider-neutral to the extent that the system runs against PostgreSQL in production and against a file-based engine for development and test without source change.

Usability

ID	Requirement and acceptance criterion
NFR-U1	The system shall meet WCAG 2.1 level AA for all member-facing pages. Criterion: automated audit clean, plus manual keyboard and contrast checks recorded
NFR-U2	Registration shall be completable on a handset in a single sitting by a member without assistance. Criterion: task completion rate and time reported
NFR-U3	The system shall obtain a System Usability Scale score of at least 68, the published average, from member participants.
NFR-U4	Every error presented to a user shall state what happened and what to do next, and shall not expose an internal identifier, stack trace or provider message.
NFR-U5	All member-facing text shall support Bengali content and shall render Bengali script correctly in all clients, including generated documents.

Reliability

ID	Requirement and acceptance criterion
NFR-R1	Monthly availability of the production service shall be at least 99 percent, excluding announced maintenance.

ID	Requirement and acceptance criterion
NFR-R2	A daily backup of the production database shall be taken and its restoration verified at least quarterly. Criterion: restoration drill recorded
NFR-R3	No single user action shall be capable of destroying a membership or financial record; archival deletion per FR-44 shall be the only deletion available through the interface.
NFR-R4	A failed external dependency, being mail delivery or file storage, shall degrade the affected function only and shall not prevent authentication or read access.
NFR-R5	A vote cast by an eligible voter shall be recorded at most once, even where the casting request is submitted more than once for the same voter and election. Criterion: the whole ballot is written in one serializable transaction that first checks the voter has no seat-vote row, and a unique index on the seat-vote row stops a second write that races past that check, so a second submission for the same voter is rejected with an already-voted error; verified in `ElectionServiceTests.CastBallotAsync_RejectsReplayAfterTheFirstVote`, which exercises a first-then-second sequential resubmission rather than two literally simultaneous requests.
NFR-R6	The mobile client shall retain a rotating on-device log of recent API errors and notable app events, so that a crash or failed request can be diagnosed from the device that produced it. Criterion: the log is capped at 256 KB, dropping the oldest half of its content once the cap is crossed rather than truncating silently, and its tail is attached to the in-app error report sent to the administrator.

Security

ID	Requirement and acceptance criterion
NFR-S1	The system shall satisfy OWASP ASVS level 2 for the control families applicable to it, with every non-conformance recorded rather than omitted.
NFR-S2	Passwords shall be stored using an adaptive hash with a cost parameter reviewed annually.
NFR-S3	Authentication attempts shall be limited to 5 per minute per source, registration to 10 per 5 minutes, and general API access to 100 per minute.
NFR-S4	National identity number, mobile number and email address shall each be unique across members, enforced in the database and not only in application code.
NFR-S5	A change of a member's credentials, roles or status shall invalidate all outstanding sessions for that member within one request.
NFR-S6	No resource shall be retrievable by a user not entitled to it by role or ownership. Criterion: the authorisation test suite exercises each protected endpoint with an unentitled principal.
NFR-S7	Uploaded files shall be validated by declared type, actual content signature and size, stored outside the web root, and served only through an authorising endpoint.
NFR-S8	Member-supplied rich text shall be sanitised on the server before storage.
NFR-S9	A member's request for his own communication history shall be scoped to that member's identity as carried in his own authenticated token, never by a client-supplied member identifier, and the equivalent view of another member's history shall be reachable only under the administrative authorisation policy.

Maintainability

ID	Requirement and acceptance criterion
NFR-M1	The dependency rule shall hold: no compile-time dependency shall point from an inner layer to an outer one. Criterion: architecture test fails the build on violation.
NFR-M2	A routine change, such as the addition of a field to an existing entity end to end, shall be completable by the maintainer within one working session.
NFR-M3	The automated suite shall run to completion in under 10 minutes on the integration runner, so that it is actually run.

ID	Requirement and acceptance criterion
NFR-M4	The system shall be operable by one maintainer with no on-call obligation, and shall require no scheduled manual intervention more often than monthly.

Portability

ID	Requirement and acceptance criterion
NFR-Po1	The system shall be deployable to any container host, with no dependency on a proprietary platform service that lacks a substitutable equivalent.
NFR-Po2	A complete environment shall be reproducible from the repository and a documented set of configuration values, with no undocumented manual step.
NFR-Po3	The Association's data shall be exportable in full in an open format on demand.

Quality-attribute scenarios

Each scenario names a source, stimulus, artefact, environment, response and response measure, so the requirement it tests can be checked.

ID	NFR	Source	Stimulus	Artefact	Environment	Response	Response measure
QAS-01	NFR-P1	Member	Opens the directory listing	API and database	Production, 50 concurrent users	Returns the first page	≤ 500 ms at p95
QAS-02	NFR-P3	Prospective member	Requests the public home page on a mid-range handset over 3G	Web client	Cold cache	Page becomes interactive	≤ 3 s; ≤ 500 KB transferred
QAS-03	NFR-S5	Administrator	Terminates a membership while that member holds a valid access token	API, session store	Normal operation	The member's next request is rejected	Within one request; no successful authenticated call after the act
QAS-04	NFR-S6	Authenticated member	Requests another member's payment record by identifier	API authorisation	Normal operation	Access refused, attempt recorded	0 unauthorised disclosures across the enumerated protected endpoints
QAS-05	NFR-R4	Mail provider	Becomes unavailable for 30 minutes	Notification component	Production	Authentication and reads continue; queued mail is retried or reported	0 failed sign-ins attributable to the outage
QAS-06	NFR-M2	Maintainer	Adds one field to an existing entity through to both clients	Whole stack	Development	Change completed with the suite green	≤ 1 working session; file-touch count reported
QAS-07	NFR-M1	Maintainer	Introduces a dependency from the domain layer onto the persistence layer	Build	Continuous integration	Build fails with the violation named	Detected on the first run, before merge

ID	NFR	Source	Stimulus	Artefact	Environment	Response	Response measure
QAS-08	NFR-F2	Examiner	Asks which code enforces the constitutional restriction on who may vote	Source and specification	Review	A single traced location is produced	Trace from Art. III §B to its domain constraint to FR-36 to a named test
QAS-09	NFR-R2	Maintainer	Loses the production database	Backup and restore procedure	Disaster	Service restored from backup	≤ 4 hours; ≤ 24 hours of data lost
QAS-10	NFR-U1	Member using a screen reader	Completes registration	Web client	Assistive technology	Registration completed unaided	No level AA violation on the registration route
QAS-11	NFR-R5	Voter	Resubmits a vote for an election already voted in	Election API, voter roll	Production	Second submission is rejected; the first vote stands unchanged	0 ballots with more than one recorded vote per voter per election